

Agent layer — the LLM consumers

Claude Code CLI · Claude Desktop (mcp-remote shim) + agx_gearb.py headless driver — non-LLM, DRAFT-only

MCP JSON-RPC tools/call

Transport + auth boundary

stdio (default) · streamable HTTP :8765 /mcp
Bearer token, constant-time compare · fail-closed boot

MCP server core — the enforcement spine · pattern: CUSTOM MCP SERVER

Trinity Loop + DFIR Swarm · trinity/ agents/ 13 deterministic SwarmAgents — no LLM inside
Architect plan → Blackboard quorum → Critic halt

FastMCP app · fastmcp_app.py
71 canonical tools (72 live) · per-tool rate limit
_safe_tool envelope — errors never crash the boundary

Thymus policy · thymus_policy.py
read-only evidence allowlist, checked before every tool exec

authenticated requests only

direct Python import (not network MCP)

findings tools

check_read gate

dispatch

IndexerClient · HTTPS

Output pipeline — findings to courtroom

record_finding

every ingest force-stamped DRAFT — the model cannot self-approve

DRAFT awaits examiner

Examiner Approval Portal :8800

HUMAN HMAC HARD-STOP

60s single-use nonce · PBKDF2-600k key

APPROVED

report_generate / report_export

APPROVED-only · HMAC-sealed
bound to evidence_image_sha256

optional, operator-gated

Wazuh SIEM push —

operator-gated
egt_single-use token + dry-run kill switch
+ false-positive denylists

Evidence — read-only

/cases E01 disk images · memory dumps
triage archives · YARA rule packs

read-only file I/O
Thymus-allowlisted

Forensic tool layer — SIFT Workstation

11 tool families · mcp_server/wrappers/
case · disk imaging · memory · registry/execution · event log
mail/maldoc · carving · findings/IOC · approval · indexer · SIEM

16 SIFT binaries + EZ-Tools
vol3 · plaso · tsik (fls/mmls/icat) · libewf · yara · bulk_extractor
RegRipper · MFTECmd/RECmd/EvtbECmd (dotnet)
argv-only subprocess — never a shell

subprocess exec

PROMPT-BASED guardrails — conventions

honest-negatives discipline · SIMULATED-approval labelling
AI-disclosure · LLM narrative rendering
each one backstopped by a code-side control

ARCHITECTURAL guardrails — enforced in code

Thymus read-only · fail-closed Bearer auth · force-DRAFT gate
HMAC + anti-replay nonce · egt_mutation tokens + kill switch
courtroom sealing + raw_stdout_sha256 provenance · APPROVED-only reports

DRAFT / APPROVED docs per-case hash chain

Wazuh Indexer (OpenSearch)

cases · evidence · findings · timeline
iocs · approvals (per-day, hash-chained)